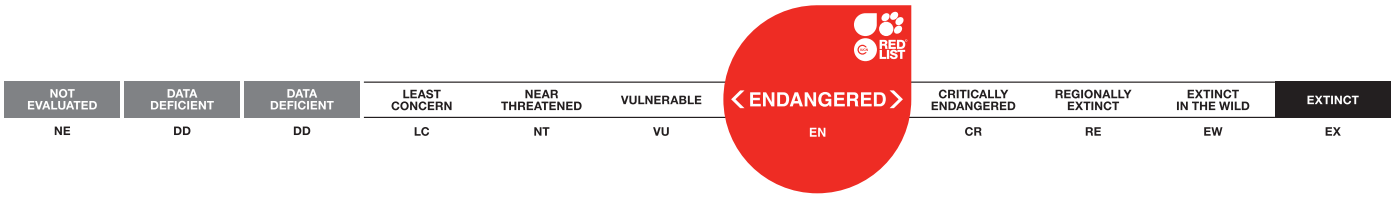
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Scope(s): Europe
Language: English

Fratercula arctica, Atlantic Puffin

Assessment by: BirdLife International



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Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Aves	Charadriiformes	Alcidae

Scientific Name: *Fratercula arctica* (Linnaeus, 1758)

Common Name(s):

- English: Atlantic Puffin, Puffin

Taxonomic Source(s):

del Hoyo, J., Collar, N.J., Christie, D.A., Elliott, A. and Fishpool, L.D.C. 2014. *HBW and BirdLife International Illustrated Checklist of the Birds of the World. Volume 1: Non-passerines*. Lynx Edicions BirdLife International, Barcelona, Spain and Cambridge, UK.

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Assessment Information

Red List Category & Criteria: Endangered A2abcde+4abcde ver 3.1

Year Published: 2021

Date Assessed: December 18 2020

Justification:

European regional assessment: Endangered (EN)
EU28 regional assessment: Least Concern (LC)

In Europe, this species has an extremely large range (its extent of occurrence (EOO) is much larger than 20,000 km² and its area of occupancy (AOO) is much larger than 2,000 km²), and hence it does not approach the thresholds for Vulnerable under the range size criteria (criteria B and D2). The population size is extremely large, (much larger than 10,000 mature individuals), and hence it does not approach the thresholds for Vulnerable under the population size criteria (criteria C and D). The probability of extinction has not been calculated for this species, therefore criterion E cannot be applied. The population trend appears to be decreasing at a very rapid rate which meets the thresholds for Endangered (EN) under the population size reduction criterion (criterion A). The species is therefore assessed as such in Europe. There is not considered to be significant potential for rescue from outside the region, and therefore the final category remains unchanged.

In the EU28, this species has a very large range (its extent of occurrence (EOO) is much larger than 20,000 km² and its area of occupancy (AOO) is much larger than 2,000 km²), and hence it does not approach the thresholds for Vulnerable under the range size criteria (criteria B and D2). The population size is extremely large, (much larger than 10,000 mature individuals), and hence it does not approach the thresholds for Vulnerable under the population size criteria (criteria C and D). The population trend appears to be increasing, and hence the species does not approach the thresholds for Vulnerable under the population size reduction criterion (criterion A). The probability of extinction has not been calculated for this species, therefore criterion E cannot be applied. For these reasons the species is evaluated as Least Concern in the EU28.

Regional Assessments:

Global

Previously Published Red List Assessments

2015 – Endangered (EN)

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Geographic Range

Range Description:

The species can be found throughout the North Atlantic Ocean. In Europe, it occurs in north-west Greenland (to Denmark), and from north Norway down to the Canary Islands, Spain in the east (Nettleship *et al.* 2014), and breeds primarily in Norway and Iceland, and also notably in the United Kingdom and the Faroe Islands.

Country Occurrence:

Native, Extant (resident): Faroe Islands; Greenland; Iceland; Ireland

Native, Extant (breeding): Norway; Russian Federation (European Russia); Svalbard and Jan Mayen; United Kingdom

Native, Extant (non-breeding): Belgium; Germany; Italy; Netherlands; Portugal; Spain; Sweden

Native, Extant (resident): Denmark; France; Gibraltar

Extant & Vagrant (seasonality uncertain): Montenegro; Serbia

Extant & Vagrant: Austria; Croatia; Finland; Hungary; Malta; Poland

Extant & Origin Uncertain (seasonality uncertain): Estonia; Latvia; Lithuania

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Distribution Map



Tiles (C) Esri -- Esri, DeLorme, NAVTEQ, TomTom, Intermap, iPC, USGS, FAO, NPS, NRCAN, GeoBase, Kadaster NL, Ordnance Survey, Esri Japan, METI, Esri China (Hong Kong), and the GIS User Community

Compiled by: BirdLife International and Handbook of the Birds of the World (2020) 2016

EXTANT (BREEDING)

EXTANT (NON-BREEDING)





The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

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Population

The European breeding population is estimated at 3,700,000-4,120,000 pairs, which equates to 7,400,000-8,240,000 mature individuals. The breeding population in the EU28 is estimated at 601,000-602,000 pairs, which equates to 1,200,000-1,210,000 mature individuals. For details of national estimates, see the Supplementary Information.

Trend Justification

In Europe, the population size is estimated to have decreased by approximately 68% (best estimate) over the past 51 years (three generations) and is expected to decline at a similar rate between 1970-2021. In the EU28, the population size is estimated to be increasing. These trends were calculated using the IUCN Criterion A tool, and based on Náttúrufræðistofnun Íslands 2018 and Fauchald *et al.* 2015, and on the national population figures provided in the Supplementary Information (see this document for details of national estimates).

For further information about this species, see Supplementary Material.

Current Population Trend: Decreasing

Habitat and Ecology

(see Appendix for additional information)

The breeding range is restricted to colder parts of the Atlantic and Arctic Oceans, with its southernmost colonies in Brittany (France). Breeding colonies are located in Iceland, Norway, Faroe Islands, the U.K., Ireland and France on islands or high cliffs.

The species nests on grassy maritime slopes, sea cliffs and rocky slopes (Nettleship *et al.* 2014). During the winter the species is highly pelagic and is dispersed widely across the sea from the Azores to the western Mediterranean and Canary Islands. When feeding chicks, birds generally forage within 10 km of their colony, but may range as far as 50 to 100 km or more (Harris 1984, Rodway and Montevecchi 1996). Birds of this species are pursuit-divers that catch most of their prey within 30 m of the water surface (Piatt and Nettleship 1985). They prey on 'forage' species, including juvenile pelagic fishes, such as herring (*Clupea harengus*), juvenile and adult capelin (*Mallotus villosus*), and sand eel (*Ammodytes spp.*) (Barrett *et al.* 1987). At times, they also prey on juvenile demersal fishes, such as gadids (Harris and Hislop 1978, Martin 1989, Rodway and Montevecchi 1996). Sand eels usually form the majority of the prey fed to chicks (Corkhill 1973, Hislop and Harris 1985, Harris and Wanless 1986, Harris and Riddiford 1989, Martin 1989), and many chicks starve during periods of low sand eel abundance (Martin 1989).

Although the generation length for both EU and Europe regional assessments were calculated using the same methodology, new information arriving after the EU assessments were undertaken gave rise to an update in the generation lengths. This new information was then used for the Europe level assessments giving rise to a difference between the generation lengths used for the EU and Europe regions.

Systems: Terrestrial, Marine

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Threats

(see Appendix for additional information)

This species is highly susceptible to the impacts of climate change, such as sea temperature rise and shifts in prey distribution and abundance (Durant *et al.* 2003, Sandvik *et al.* 2005). This is a particularly important threat when prey species are exploited unsustainably, leading to prey reductions and subsequent unsuccessful breeding. The species is vulnerable to oil spills and other marine pollution. The species is also vulnerable to extreme weather events and storms, with large wrecks recorded following severe winter storms at sea. At the breeding colonies the species is vulnerable to invasive predators, such as rats, cats, and American Mink (*Neovison vison*). The species is susceptible to being caught in gillnets, although other fishing gears may also catch significant numbers. Increasing numbers of offshore wind farms may result in displacement from habitat, although the risk of collision is considered very low (Bradbury *et al.* 2014). The species is hunted for human consumption in Iceland, and in the Faroe Islands (Thorup *et al.* 2014).

Conservation Actions

(see Appendix for additional information)

Conservation Actions Underway

The species is listed under the African Eurasian Waterbird Agreement. The species is included in the Action plan for seabirds in Western-Nordic areas (2010). There are 76 marine Important Bird Areas identified across the European region. Within the EU there are 40 Special Protection Areas which list this species as occurring within its boundaries.

Conservation Actions Proposed

Further identification of important sites for this species, particularly in offshore regions and designation as marine protected areas; Identify the risks of different activities on seabirds, and locations sensitive to seabirds. Continue eradication of invasive predators from breeding colonies. Management of fisheries to ensure long term sustainability of key stocks (e.g. sand eels). Establish observer schemes for bycatch and prepare National/European Community plans of action on seabird bycatch. Continue AMAP monitoring of seabird contaminants; include new contaminants and secure communication between seabird and contaminants research. Increase the level of understanding among the public of introducing hunting restrictions. Develop codes-of-conduct for more organised activities (e.g. tourism). Ensure that appropriate protection (national laws and international agreements) applies to new areas and times in cases of changes in seabird migration routes and times.

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Credits

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Appendix

For more information on IUCN Classification Schemes referenced in this appendix, please visit: http://www.iucnredlist.org/technical-documents/classification-schemes

Habitats

Habitat	Season	Suitability	Major Importance?
4. Grassland -> 4.2. Grassland - Subarctic	Breeding Season	Suitable	Yes
4. Grassland -> 4.4. Grassland - Temperate	Breeding Season	Suitable	Yes
9. Marine Neritic -> 9.1. Marine Neritic - Pelagic	Breeding Season	Suitable	Yes
9. Marine Neritic -> 9.1. Marine Neritic - Pelagic	Non-Breeding Season	Suitable	Yes
9. Marine Neritic -> 9.2. Marine Neritic - Subtidal Rock and Rocky Reefs	Breeding Season	Suitable	Yes
9. Marine Neritic -> 9.2. Marine Neritic - Subtidal Rock and Rocky Reefs	Non-Breeding Season	Suitable	Yes
9. Marine Neritic -> 9.3. Marine Neritic - Subtidal Loose Rock/pebble/gravel	Breeding Season	Suitable	Yes
9. Marine Neritic -> 9.3. Marine Neritic - Subtidal Loose Rock/pebble/gravel	Non-Breeding Season	Suitable	Yes
9. Marine Neritic -> 9.4. Marine Neritic - Subtidal Sandy	Breeding Season	Suitable	Yes
9. Marine Neritic -> 9.4. Marine Neritic - Subtidal Sandy	Non-Breeding Season	Suitable	Yes
9. Marine Neritic -> 9.5. Marine Neritic - Subtidal Sandy-Mud	Breeding Season	Suitable	Yes
9. Marine Neritic -> 9.5. Marine Neritic - Subtidal Sandy-Mud	Non-Breeding Season	Suitable	Yes
9. Marine Neritic -> 9.7. Marine Neritic - Macroalgal/Kelp	Non-Breeding Season	Suitable	Yes
9. Marine Neritic -> 9.7. Marine Neritic - Macroalgal/Kelp	Breeding Season	Suitable	Yes
9. Marine Neritic -> 9.9. Marine Neritic - Seagrass (Submerged)	Non-Breeding Season	Suitable	Yes
9. Marine Neritic -> 9.9. Marine Neritic - Seagrass (Submerged)	Breeding Season	Suitable	Yes
10. Marine Oceanic -> 10.1. Marine Oceanic - Epipelagic (0-200m)	Breeding Season	Suitable	No
10. Marine Oceanic -> 10.1. Marine Oceanic - Epipelagic (0-200m)	Non-Breeding Season	Suitable	No
10. Marine Oceanic -> 10.2. Marine Oceanic - Mesopelagic (200-1000m)	Non-Breeding Season	Suitable	No

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Habitats (Continued)

Habitat	Season	Suitability	Major Importance?
10. Marine Oceanic -> 10.2. Marine Oceanic - Mesopelagic (200-1000m)	Breeding Season	Suitable	No
12. Marine Intertidal -> 12.1. Marine Intertidal - Rocky Shoreline	Non-Breeding Season	Suitable	No
12. Marine Intertidal -> 12.1. Marine Intertidal - Rocky Shoreline	Breeding Season	Suitable	No
12. Marine Intertidal -> 12.6. Marine Intertidal - Tidepools	Non-Breeding Season	Suitable	No
12. Marine Intertidal -> 12.6. Marine Intertidal - Tidepools	Breeding Season	Suitable	No
13. Marine Coastal/Supratidal -> 13.1. Marine Coastal/Supratidal - Sea Cliffs and Rocky Offshore Islands	Breeding Season	Suitable	No

Use and Trade

End Use	Local	National	International
1. Food - human	No	Yes	No
13. Pets/display animals, horticulture	No	No	Yes

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Threats

Threat	Timing	Scope	Severity
3. Energy production & mining -> 3.3. Renewable energy **Stresses:** 1. Ecosystem stresses -> 1.3. Indirect ecosystem effects 2. Species Stresses -> 2.1. Species mortality 2. Species Stresses -> 2.2. Species disturbance	Ongoing	Minority (<50%)	Causing/Could cause fluctuations
5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.1. Intentional use (species is the target) **Stresses:** 2. Species Stresses -> 2.1. Species mortality	Ongoing	Minority (<50%)	Slow, Significant Declines
5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.4. Unintentional effects: (large scale) [harvest] **Stresses:** 2. Species Stresses -> 2.1. Species mortality	Ongoing	Unknown	Rapid Declines
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Neogale vison) **Stresses:** 2. Species Stresses -> 2.1. Species mortality 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success	Ongoing	Majority (50-90%)	Slow, Significant Declines
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Unspecified Rattus) **Stresses:** 2. Species Stresses -> 2.1. Species mortality 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success	Ongoing	Majority (50-90%)	Slow, Significant Declines
9. Pollution -> 9.2. Industrial & military effluents -> 9.2.1. Oil spills **Stresses:** 1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.1. Species mortality	Past, Likely to Return	Unknown	Rapid Declines
9. Pollution -> 9.2. Industrial & military effluents -> 9.2.3. Type Unknown/Unrecorded **Stresses:** 2. Species Stresses -> 2.1. Species mortality	Ongoing	Unknown	Unknown
11. Climate change & severe weather -> 11.1. Habitat shifting & alteration **Stresses:** 1. Ecosystem stresses -> 1.2. Ecosystem degradation 1. Ecosystem stresses -> 1.3. Indirect ecosystem effects	Ongoing	Unknown	Unknown

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Threats (Continued)

Threat	Timing	Scope	Severity
11. Climate change & severe weather -> 11.3. Temperature extremes **Stresses:** 1. Ecosystem stresses -> 1.3. Indirect ecosystem effects 2. Species Stresses -> 2.2. Species disturbance	Ongoing	Unknown	Unknown
11. Climate change & severe weather -> 11.4. Storms & flooding **Stresses:** 1. Ecosystem stresses -> 1.3. Indirect ecosystem effects 2. Species Stresses -> 2.1. Species mortality	Ongoing	Unknown	Rapid Declines
11. Climate change & severe weather -> 11.5. Other impacts **Stresses:** 1. Ecosystem stresses -> 1.2. Ecosystem degradation 1. Ecosystem stresses -> 1.3. Indirect ecosystem effects	Ongoing	Unknown	Unknown

Conservation Actions in Place

Conservation Action in Place
In-place research and monitoring
Action Recovery Plan: Yes
Systematic monitoring scheme: No
In-place land/water protection
Conservation sites identified: Yes, over entire range
Occurs in at least one protected area: Yes
Invasive species control or prevention: No
In-place species management
Successfully reintroduced or introduced benignly: No
Subject to ex-situ conservation: No
In-place education
Subject to recent education and awareness programmes: No
Included in international legislation: Yes
Subject to any international management / trade controls: No

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Conservation Actions Needed

Conservation Action Needed	Notes
1. Land/water protection -> 1.1. Site/area protection	-
1. Land/water protection -> 1.2. Resource & habitat protection	-
2. Land/water management -> 2.1. Site/area management	-
2. Land/water management -> 2.2. Invasive/problematic species control	-
4. Education & awareness -> 4.3. Awareness & communications	-
5. Law & policy -> 5.2. Policies and regulations	-
5. Law & policy -> 5.4. Compliance and enforcement -> 5.4.4. Scale unspecified	-

Research Needed

Research Needed	Notes
1. Research -> 1.5. Threats	-
3. Monitoring -> 3.1. Population trends	-

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Additional Data Fields

Distribution
Continuing decline in area of occupancy (AOO): Unknown
Extreme fluctuations in area of occupancy (AOO): No
Continuing decline in extent of occurrence (EOO): Unknown
Extreme fluctuations in extent of occurrence (EOO): No

Population
Number of mature individuals: 7400000-8240000,7820000
Continuing decline of mature individuals: Yes
Extreme fluctuations: No
Population severely fragmented: No
Continuing decline in subpopulations: Unknown
Extreme fluctuations in subpopulations: No
All individuals in one subpopulation: No
Percent Change In Past: 68
Reduction Or Increase In Past: Reduction
Past Population Reduction Basis: a) direct observation; b) an index of abundance appropriate for the taxon; c) a decline in area of occupancy, extent of occurrence and/or quality of habitat; d) actual or potential levels of exploitation; e) the effects of introduced taxa, hybridization, pathogens, pollutants, competitors or parasites)
Past Causes Of Reduction: Reversible: ; Understood: ; Ceased: No
Ongoing Percent Change: 68
Ongoing Reduction Or Increase: Reduction
Ongoing Population Reduction Basis: a) direct observation; b) an index of abundance appropriate for the taxon; c) a decline in area of occupancy, extent of occurrence and/or quality of habitat; d) actual or potential levels of exploitation; e) the effects of introduced taxa,hybridization, pathogens, pollutants,competitors or parasites
Ongoing Causes Of Reduction: Reversible: Unknown; Understood: Yes; Ceased: No

Habitats and Ecology
Continuing decline in area, extent and/or quality of habitat: Unknown
Generation Length (years): 17
Movement patterns: Full Migrant
Congregatory: Congregatory (and dispersive)

Additional Data Fields (Continued)

Population
Number of mature individuals: 7400000-8240000,7820000
Continuing decline of mature individuals: Yes
Extreme fluctuations: No
Population severely fragmented: No
Continuing decline in subpopulations: Unknown
Extreme fluctuations in subpopulations: No
All individuals in one subpopulation: No
Percent Change In Past: 68
Reduction Or Increase In Past: Reduction
Past Population Reduction Basis: a) direct observation; b) an index of abundance appropriate for the taxon; c) a decline in area of occupancy, extent of occurrence and/or quality of habitat; d) actual or potential levels of exploitation; e) the effects of introduced taxa, hybridization, pathogens, pollutants, competitors or parasites)
Past Causes Of Reduction: Reversible: ; Understood: ; Ceased: No
Ongoing Percent Change: 68
Ongoing Reduction Or Increase: Reduction
Ongoing Population Reduction Basis: a) direct observation; b) an index of abundance appropriate for the taxon; c) a decline in area of occupancy, extent of occurrence and/or quality of habitat; d) actual or potential levels of exploitation; e) the effects of introduced taxa,hybridization, pathogens, pollutants,competitors or parasites
Ongoing Causes Of Reduction: Reversible: Unknown; Understood: Yes; Ceased: No

Habitats and Ecology
Continuing decline in area, extent and/or quality of habitat: Unknown
Generation Length (years): 17
Movement patterns: Full Migrant
Congregatory: Congregatory (and dispersive)

Distribution
Continuing decline in number of locations: Unknown
Extreme fluctuations in the number of locations: No

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The IUCN Red List Partnership

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